

DATA VISUALIZATION • FINAL PROJECT • SUMMER 2026

Brazil's Marketplace in Motion

A data story of the Olist e-commerce ecosystem, 2016–2018 — 100k real orders, 27 states, one marketplace under strain.

100k+

orders analyzed

12

analytical questions

27

Brazilian states

3

linked deliverables

A rich, real-world marketplace — not a toy dataset



Numerical

Price, freight, product weight & dimensions, review scores



Categorical

72 product categories, payment types, order status



Spatial

Customer & seller state / city / zip, geolocation



Temporal

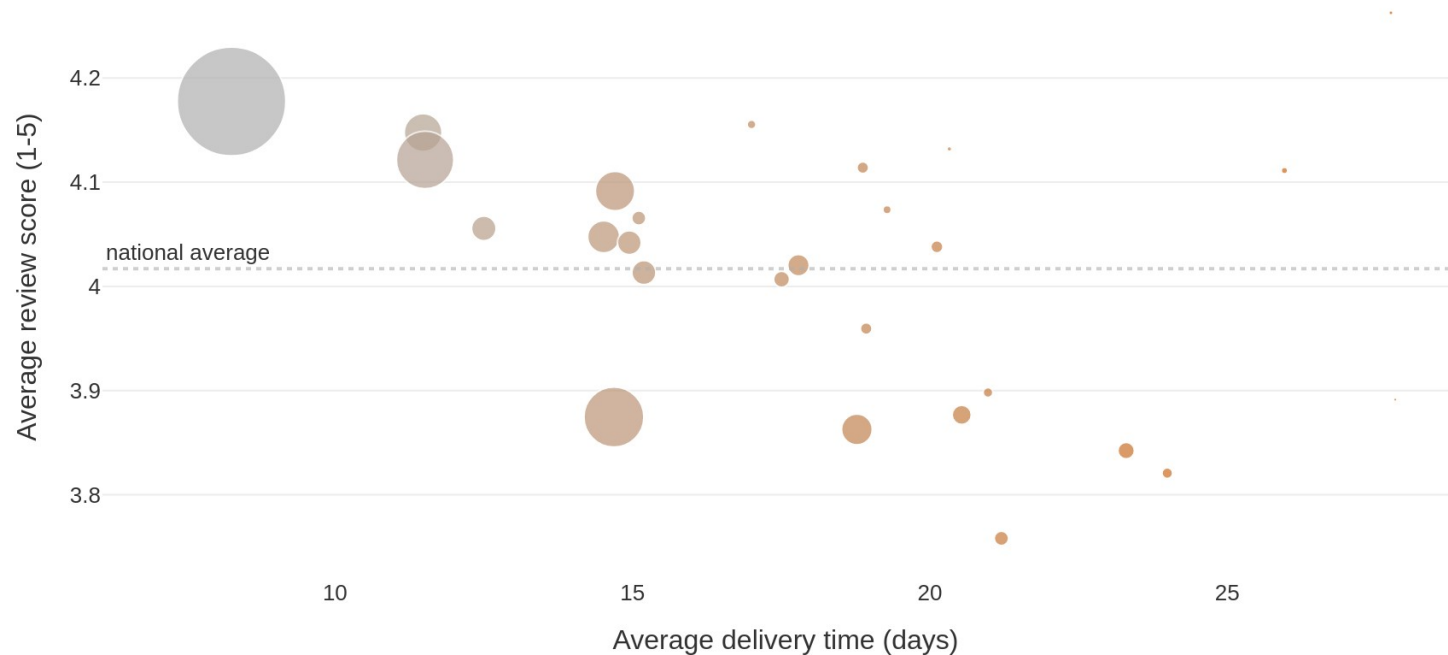
Purchase, approval, shipping & delivery timestamps, 2016–18

Source: olist/work-at-olist-data on GitHub — the official mirror of the Kaggle "Brazilian E-Commerce Public Dataset by Olist". 9 relational tables merged into a 112,650-row, 45-column order-item master table.

One dataset. Twelve questions. One story.

Slower delivery states also rate their orders lower

Slower-delivery states also rate their orders lower — the North lags on both fronts

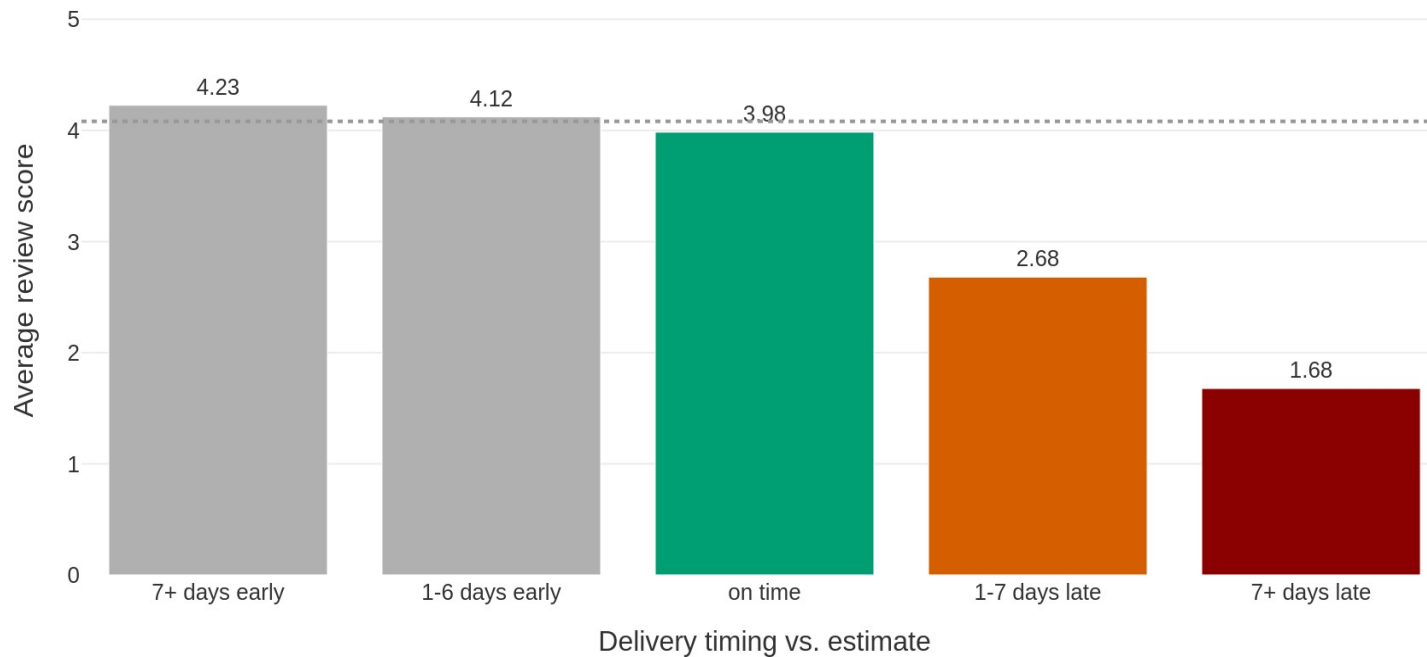


Takeaway

Northern/Northeastern states average 15–21+ day deliveries and sit visibly below the national review average — the South-East core (SP/PR/MG) delivers fastest and rates highest.

A late delivery costs almost a full star

A late delivery costs almost a full star — lateness is the single biggest review-score killer

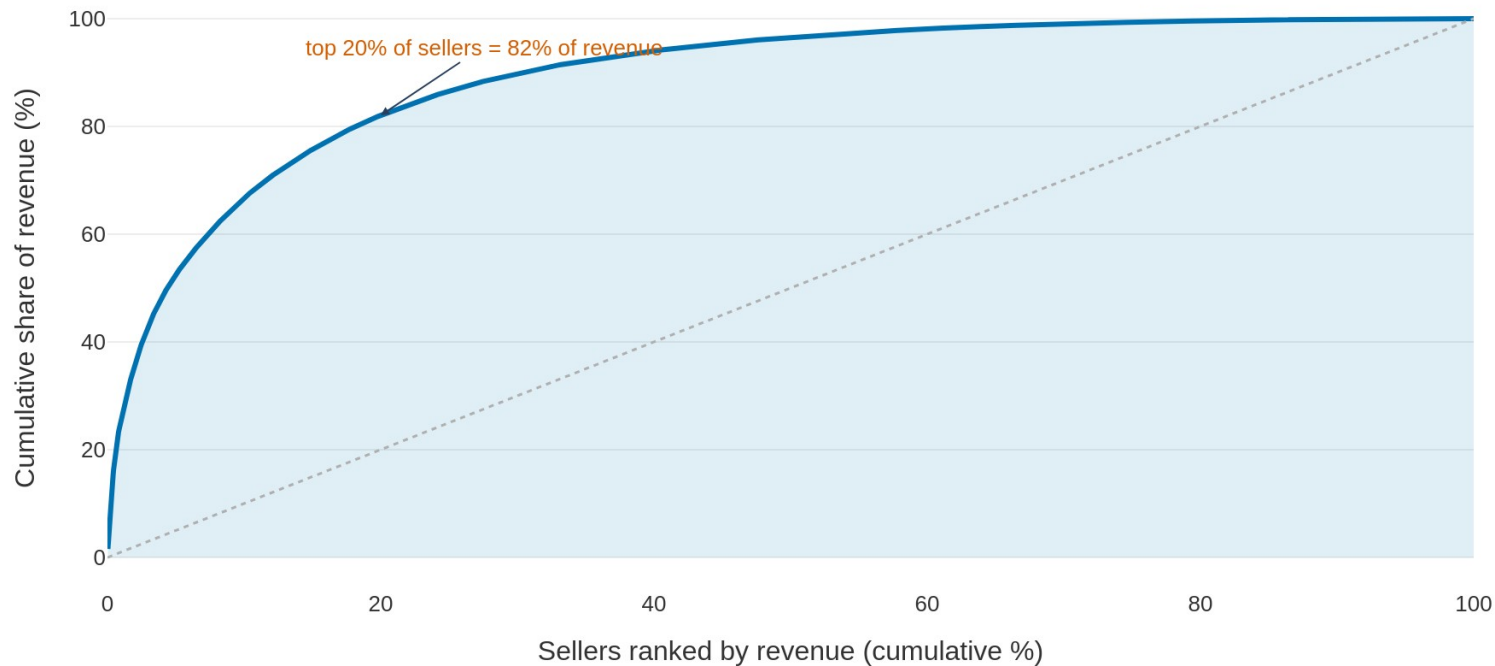


Takeaway

Average review score falls from 3.98 (on-time) to 2.68 for orders 1-7 days late, and 1.68 past a week late. Lateness — not early arrival — is the single biggest driver of review score.

Revenue is sharply concentrated among a few sellers

Revenue is sharply concentrated: a small slice of sellers accounts for most marketplace revenue

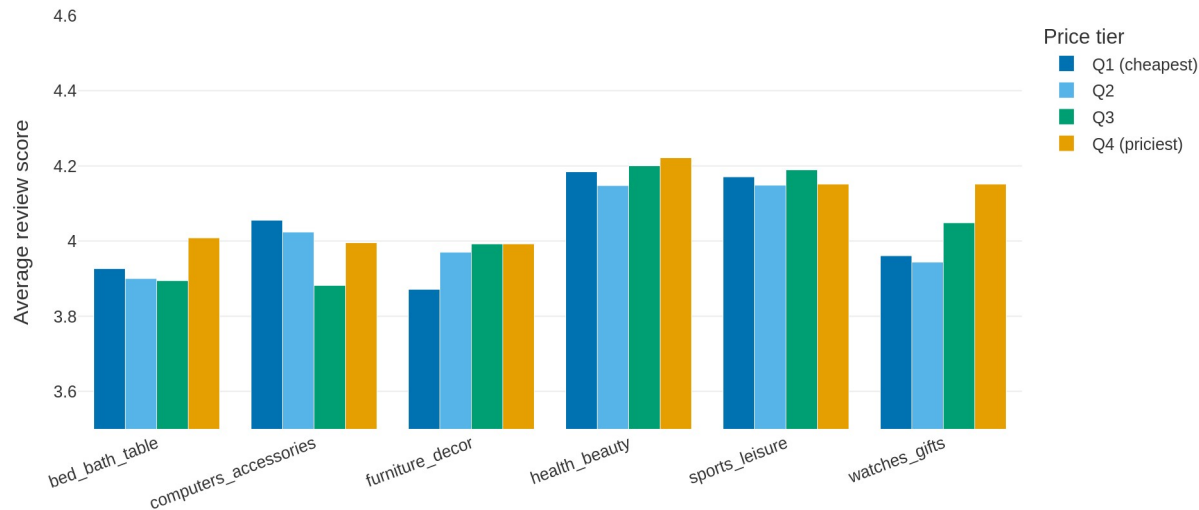


Takeaway

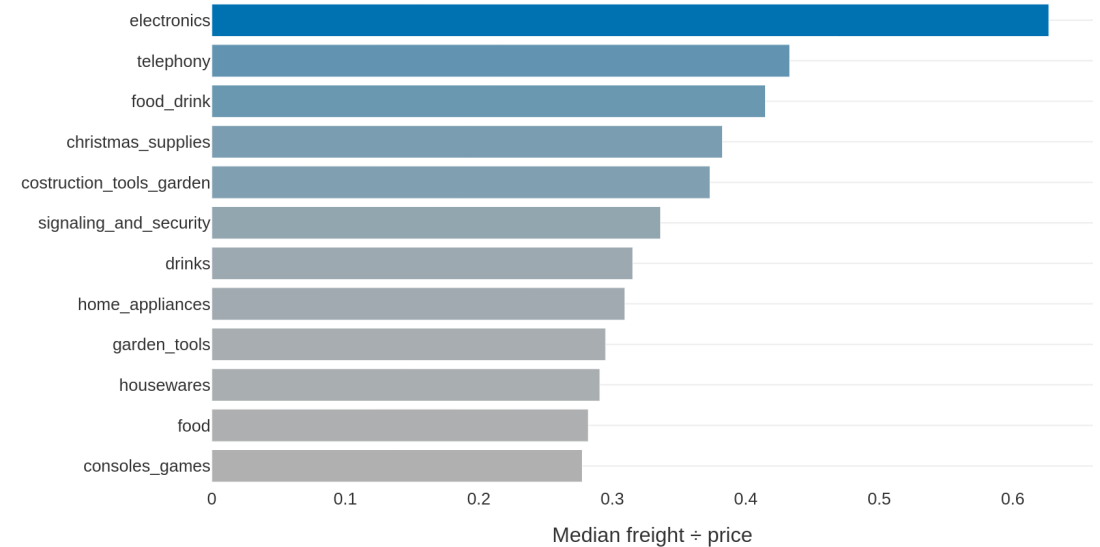
The top 20% of sellers generate roughly 82% of marketplace revenue — a classic Pareto pattern that makes seller-relationship management a high-leverage lever for Olist.

Price doesn't buy satisfaction — but size drives freight cost

Higher price tiers don't buy higher satisfaction — the pattern is flat within every top category



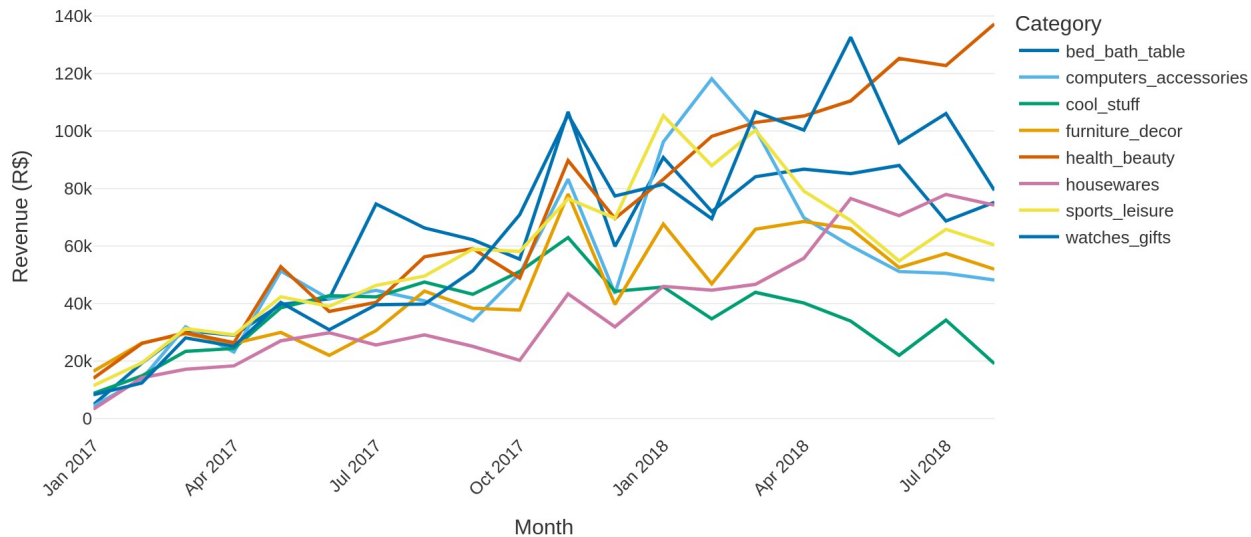
Bulky, low-value items like furniture & construction goods carry the heaviest relative freight cost



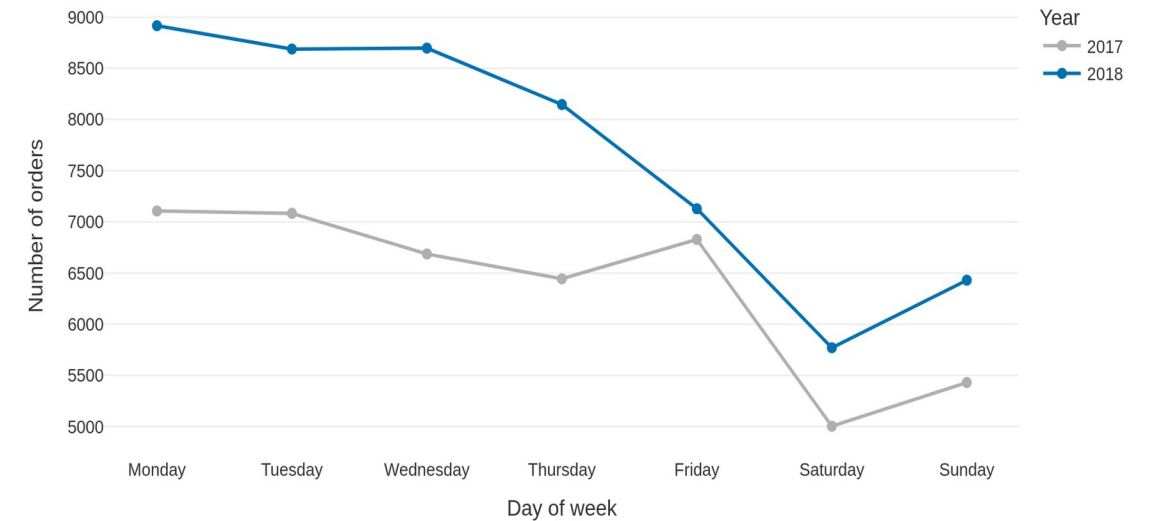
Within every top-selling category, review scores stay essentially flat from cheapest to priciest quartile — while bulky, low-value categories like furniture and construction goods carry the heaviest freight-to-price burden. Satisfaction is an operations problem, not a pricing one.

2018 growth was healthy, broad-based, and increasingly weekday-driven

Health & beauty and watches/gifts pulled ahead of the pack through 2018



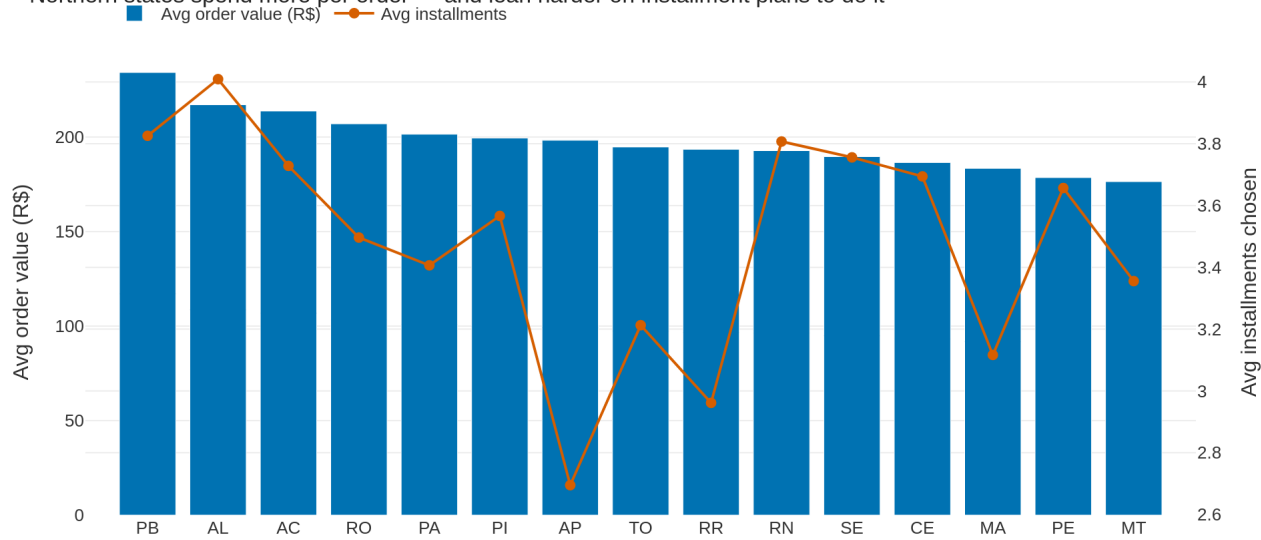
Weekday shopping habit strengthened in 2018 — weekend orders grew slower than the rest of the week



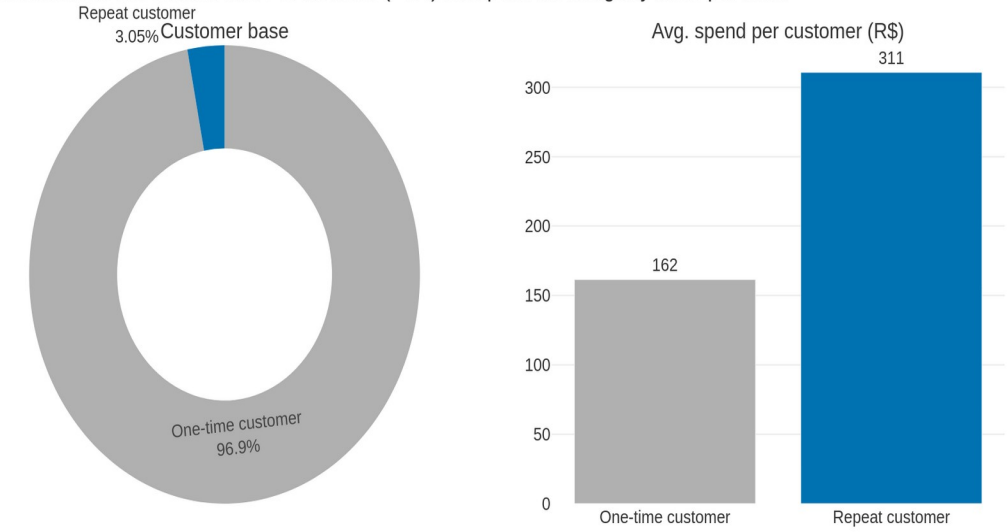
Health & beauty and watches/gifts pulled ahead of other categories through 2018, while weekday order volume grew faster than weekends year-over-year — a marketplace maturing past its early, weekend-shopping-driven phase.

Northern states spend more, finance longer, and rarely come back

Northern states spend more per order — and lean harder on installment plans to do it



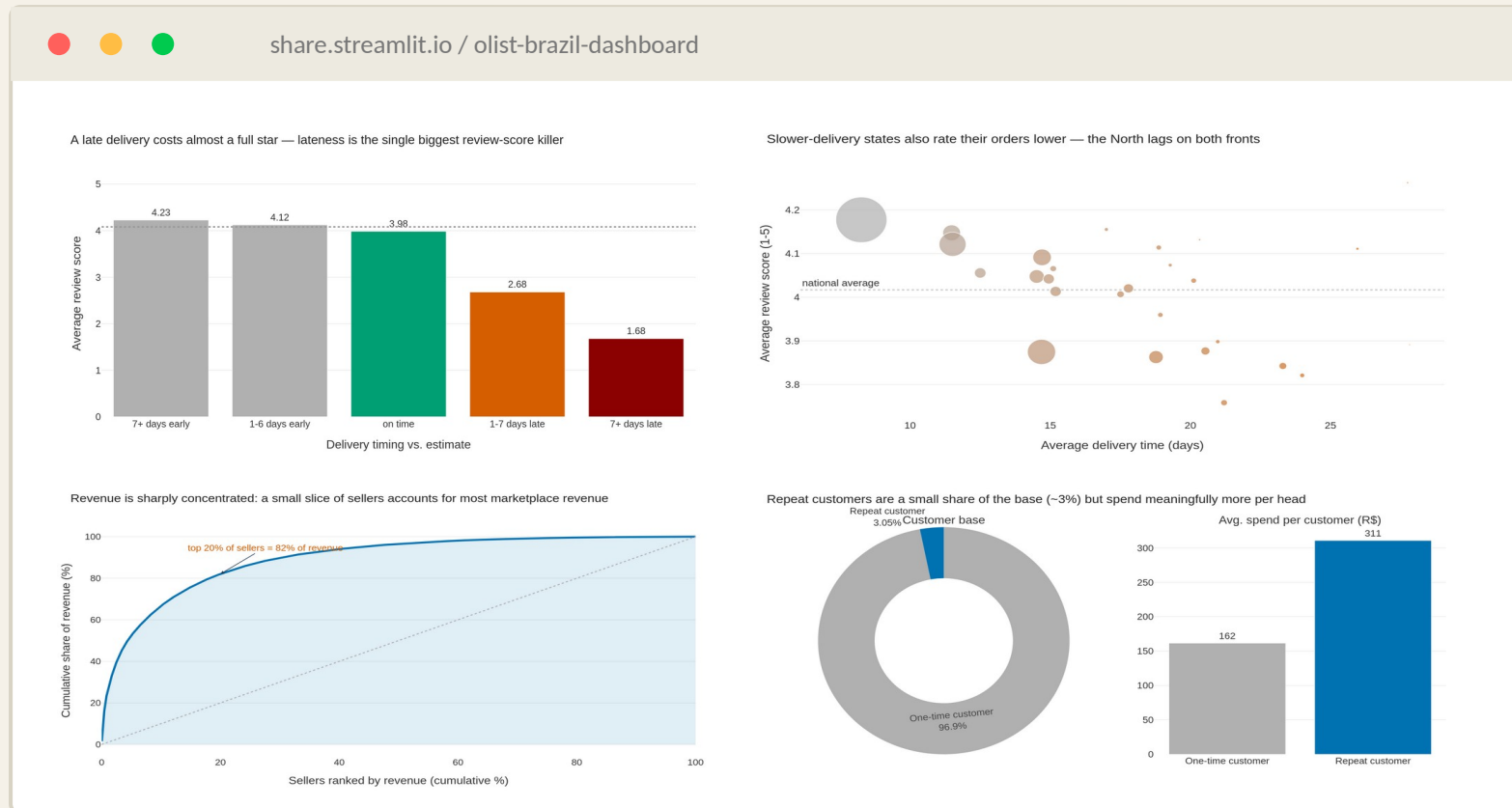
Repeat customers are a small share of the base (~3%) but spend meaningfully more per head



Northern states post the highest average order values and lean hardest on installment financing. Meanwhile repeat customers are just ~3% of the base — but spend roughly double per head — making retention a clear, under-exploited growth lever.

Explore it yourself — the live Streamlit dashboard

A curated, filterable view across three tabs — Sales & Categories, Delivery & Satisfaction, Geography & Customers — built with the same CVD-safe palette and design language as the notebook.



Tabs

Sales & Categories

Delivery & Satisfaction

Geography & Customers

Filters: year · state · product category

Live URL: [to be added on deploy]
Repo: github.com/<user>/olist-brazil-project

Conclusions

What Olist should do next

1 **Fix delivery reliability first**

It's the single strongest lever on review score — bigger than price, category, or payment method.

2 **Support outlying sellers/states operationally**

Northern & Northeastern states carry the worst delivery times and heaviest freight ratios — targeted logistics investment there compounds.

3 **Invest in seller diversification**

Top 20% of sellers drive ~82% of revenue — concentration risk worth actively managing.

4 **Build retention programs**

Repeat customers are rare (~3%) but spend ~2x more — even a modest lift in repeat rate is high-value.